MATHEMATICS

MATHEMATICS GRADE 12 TERM 3 CONTROLLED TEST

SUBJECT	MATHEMATICS
GRADE	12
ASSESSMENT TASK	MATHEMATICS GRADE 12 TERM 3 CONTROLLED TEST
MARKS	50
TIME	1.5 hours

Educator approval required before classroom use.

INSTRUCTIONS AND INFORMATION

1. Read all instructions carefully.

2. Answer all questions or complete all activities as instructed.

3. Show all working where calculations, planning, diagrams or explanations are required.

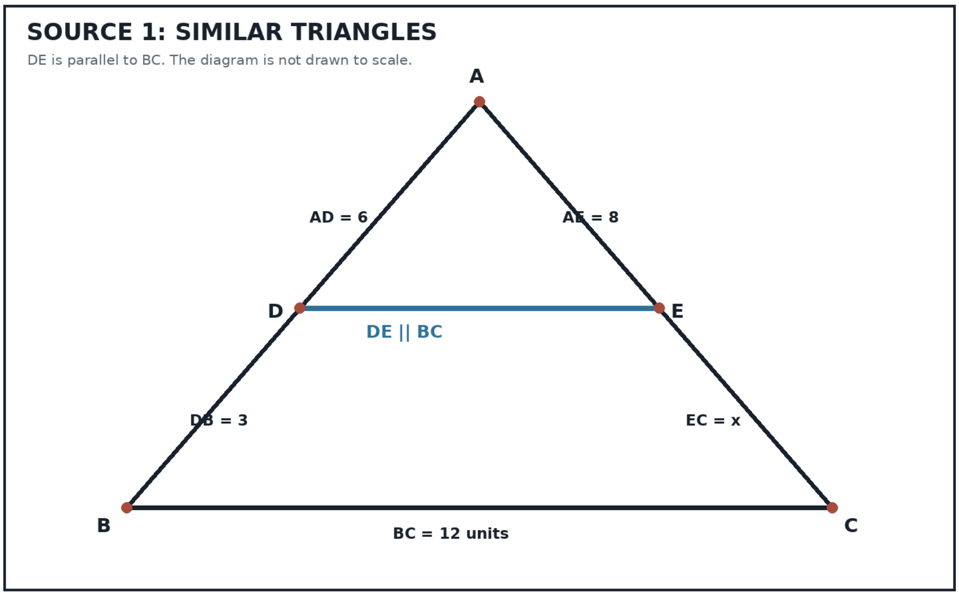
4. Use the space provided by the educator, answer booklet or learner task book.

5. Write neatly and clearly.

6. Do not use a calculator unless the educator allows it.

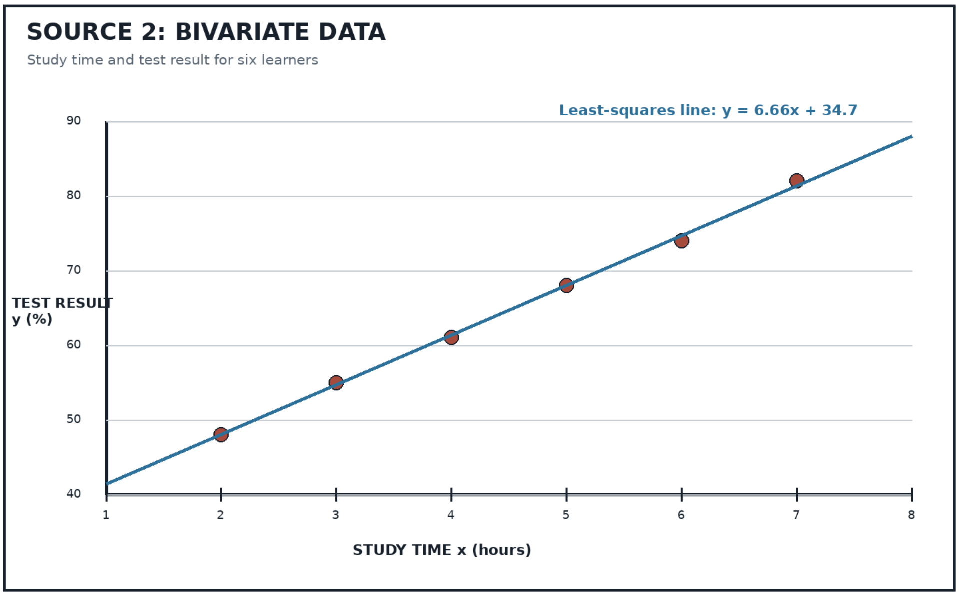
SOURCE MATERIAL

SOURCE 1: Similar triangles



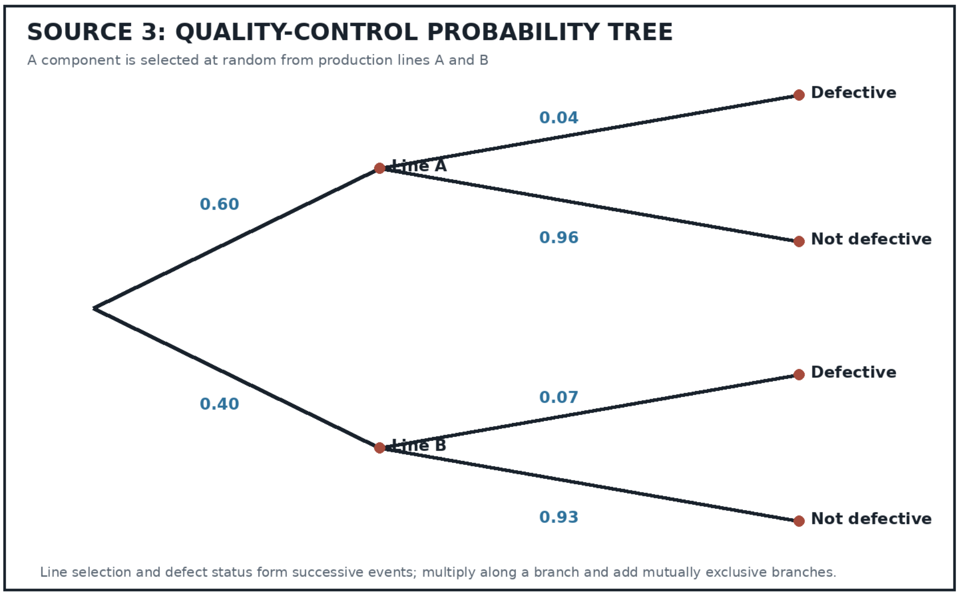
Source Type: Euclidean Geometry Diagram

SOURCE 2: Regression and correlation dataset



Source Type: Scatterplot And Regression Line

SOURCE 3: Quality-control probability tree



Source Type: Probability Tree

QUESTIONS

QUESTION 1: EUCLIDEAN GEOMETRY

Instructions: Use Source 1. Give reasons for all geometrical statements.

Stimulus: Refer to SOURCE 1.

1.1. Prove that triangle ADE is similar to triangle ABC. **(4)**

1.2. Hence prove that AD/DB = AE/EC. **(5)**

1.3. Calculate x, the length of EC. **(3)**

1.4. Calculate the length of DE. **(3)**

1.5. F is the midpoint of AB. A line through F parallel to BC meets AC at G. Prove that G is the midpoint of AC. **(5)**

QUESTION 2: REGRESSION AND CORRELATION

Instructions: Use Source 2. Round numerical answers to one decimal place where necessary.

Stimulus: Refer to SOURCE 2.

2.1. Identify the dependent variable. **(1)**

2.2. Describe the direction and strength of the correlation. **(2)**

2.3. Use the regression line to estimate the result for 6.5 hours of study. **(2)**

2.4. Is the estimate in Question 2.3 an interpolation or an extrapolation? Give a reason. **(2)**

2.5. Explain why the strong correlation does not by itself prove that extra study time causes higher marks. **(3)**

QUESTION 3: COUNTING AND PROBABILITY

Instructions: Use Source 3 where indicated. Show all probability calculations.

Stimulus: Refer to SOURCE 3 for Questions 3.1 to 3.4.

3.1. Verify that the probabilities on each pair of branches sum to 1. **(2)**

3.2. Calculate the probability that a randomly selected component is defective. **(4)**

3.3. Given that a component is defective, calculate the probability that it came from Line B. **(4)**

3.4. Determine whether the events 'Line B' and 'defective' are independent. Justify your answer. **(3)**

3.5. A batch code contains two different letters followed by three digits. Repetition of digits is allowed. How many different codes are possible? **(4)**

3.6. Three components are selected independently from a process with defect probability 0.052. Calculate the probability that at least one is defective. **(3)**